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#2026-08-24
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#Problem 1 of Assignment

.globl main
.text

main:
lw $t1, aap
lw $t2, na
lw $t3, d
mul $t4, $t3, 7
add $s0, $t4, $t1 #$s0 stores the 8th term of AP

#Directly using the formula of sum of first n terms of AP
lb $t6, sa
#div $t7, $t6, 2 #Instruction Working but not allowed
li $t9, 2
div $t6, $t9
mflo $t7
mul $t1, $t1, 2 #2a calculated
addi $t8, $t6, -1
mul $t8, $t8, $t3
add $t1, $t1, $t8
mul $s1, $t7, $t1 #$s1 stores the sum of AP

lw $t1, agp
lw $t2, ng
lw $t3, r
mul $t4, $t3, $t3
mul $t4, $t4, $t3 #r^3 calculated
mul $s3, $t4, $t2 #$s3 stores the 4th term of GP

addi $t0, $t3, -1 #calulate (r-1)
#div $t1, $t1, $t0 #calculate a/(r-1)
div $t1, $t0
mflo $t1
mul $t4, $t4, $t3 #Because we already have r^3, we may reuse it instead of calculating
it again.
addi $t4, $t4, -1
mul $s4, $t1, $t4 #$s4 now stores the result of sum of GP


li $v0, 10 # Sets $v0 to "10" to select exit syscall
syscall



.data
aap: .word 1
agp: .word 4
d: .word 10
r: .word 2
na: .word 8
ng: .word 4
sa: .byte 6
sg: .byte 4
```